

JAWAD ALI

AI Engineer

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CAREER SUMMARY

Results-driven AI/ML Engineer and BS Artificial Intelligence graduate from COMSATS University Islamabad with hands-on experience building production-ready Generative AI systems, LLM applications, and end-to-end automation pipelines. Specializes in designing agentic workflows using LangChain and LangGraph, building Retrieval-Augmented Generation (RAG) pipelines with vector databases such as FAISS and Pinecone, and deploying LLM-powered chatbots and copilots via FastAPI and Streamlit. Strong practical experience with n8n for orchestrating multi-step automation workflows, integrating third-party APIs, and building event-driven data pipelines for real business operations. Proficient in MLOps practices including experiment tracking with MLflow, containerization with Docker, CI/CD pipelines, and cloud deployments on AWS, Azure, and GCP. Beyond core engineering, has delivered full-stack web applications (MERN), freelance automation and analytics projects (n8n, Power BI), and computer vision/ML practice projects. Committed to delivering scalable, clean AI architectures that drive measurable business value.

EXPERIENCE

AI / ML Engineer

Aug 2025 – Present

OptimAIze BE (Remote, Belgium)

Responsibilities:

- Designed and deployed autonomous LangChain ReAct agents and LangGraph-based multi-agent systems, automating complex multi-step business workflows across cloud infrastructure.
- Built end-to-end AI automation workflows using n8n for intelligent scheduling, data ingestion, webhook handling, routing, and third-party API orchestration.
- Implemented and deployed Retrieval-Augmented Generation (RAG) pipelines integrating FAISS and Pinecone vector databases for scalable semantic search and document-based question answering.
- Developed production-grade deep learning and classical ML pipelines for classification, regression, and NLP tasks; integrated trained models into live systems.
- Connected and fine-tuned LLMs (OpenAI GPT-4, Claude API, HuggingFace Transformers) to CRMs, databases, and operational services.
- Built a handwritten sentence-checking tool using Tesseract OCR for text extraction combined with OpenAI for reading/comprehension enhancement, in Python.
- Built interactive demo and evaluation interfaces using Streamlit to validate LLM outputs, RAG pipeline accuracy, and end-to-end agent workflows.
- Enforced MLOps best practices including MLflow experiment tracking, Docker containerization, and CI/CD pipelines on AWS and Azure.
- Engineered prompt engineering strategies and tool/function-calling architectures to improve LLM reasoning and reduce hallucinations.
- Managed the company LinkedIn presence, posting AI industry news, company updates, and related content.
- Handled part of the HR process — posting job openings, screening and shortlisting candidate CVs, and progressing shortlisted candidates through hiring.
- Worked solo end-to-end on all technical deliverables in this role, from architecture design to deployment.

Tools & Technologies:

- Generative AI & Agents: LangChain, LangGraph, ReAct Agents, Agentic Workflows, Tool / Function Calling, Prompt Engineering, Copilot Studio
- LLMs & Models: OpenAI GPT-4 / GPT-4o, Claude API (Anthropic), HuggingFace Transformers, Sentence Transformers, Multilingual LLMs, LLM Fine-tuning
- Retrieval & Search: RAG Pipelines, Semantic Search, FAISS, Pinecone, Document Parsing, Vector Similarity Search
- MLOps & Deployment: Docker, MLflow, CI/CD Pipelines, AWS (EC2, S3, Lambda), Azure, GCP, Kubernetes, Environment Management, Logging & Monitoring
- Automation & Orchestration: n8n, Webhook Handling, Scheduled Jobs, Event-Driven Pipelines, Third-Party API Integration
- Backend & APIs: Python, FastAPI, Flask, RESTful API Design, JWT Authentication, PostgreSQL, MongoDB, Redis, MySQL

- ML & Deep Learning: CNNs, RNNs, LSTMs, Transformers, Transfer Learning, scikit-learn, XGBoost, LightGBM, CatBoost, Ensemble Methods
- Data & NLP: Pandas, NumPy, NLTK, spaCy, OCR (Tesseract), Data Preprocessing, Feature Engineering, Statistical Analysis
- Frontend & Demos: Streamlit, React.js, Node.js, Express.js, MERN Stack
- Version Control & DevOps: Git, GitHub, CI/CD Fundamentals, Branching & Merging, Code Reviews

EDUCATION

COMSATS University Islamabad *BS in Artificial Intelligence*

2022 – 2026

Relevant Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Data Science, Statistical Modelling, Neural Networks, Databases, Software Engineering

PROJECTS

Car Price Analyst Agent

2025

Stack: *Python • LangChain • OpenAI API • XGBoost • Selenium • BeautifulSoup • MongoDB • Streamlit • Cron Jobs*

- Built an end-to-end system scraping second-hand car listings from AutoScout24, GoCars, and 2ehands, scaling from an initial 20–30k listings to 200k+ listings.
- Trained and compared XGBoost plus two additional models for price prediction, reaching ~92% accuracy against a 97% target; shipped a rule-based pricing approach for the MVP while the trained ML model remains deployed live in parallel.
- Set up a cron job on a remote server to run the scraper on a schedule, clean incoming data automatically, and store it in MongoDB — a self-sustaining pipeline requiring no manual trigger.
- Delivered the user-facing product as a Streamlit app for on-demand vehicle price checks.

Business Analyser Agent

2026

Stack: *Python • LangChain ReAct • OpenAI API • Streamlit • MLflow • AWS*

- Built an autonomous agent that acts as a personal business analyst for a company, answering conversational questions about its projects and generating business proposal documents.
- Implemented a multi-step LangChain ReAct agent with tool-calling, memory management, and intelligent decision routing.
- Delivered as a Streamlit chatbot application combining conversational Q&A and document viewing in one interface.

HEZOP Analyzer Agent

2025

Stack: *Python • LangChain • OpenAI API • RAG • FastAPI • Streamlit*

- Built for a real client use case: an AI agent answering process-safety engineering questions using a RAG pipeline over safety documents.
- Converted freelancer-exported AutoCAD DXF files into structured JSON, then built the analysis and RAG pipeline on top of that data.
- Implemented a RAG-based knowledge pipeline over engineering safety documents, enabling HEZOP engineers to get answers grounded in official documentation.

Job Portal Web App

2025

Stack: *React.js • Node.js • Express.js • MongoDB • Mongoose • JWT • bcrypt*

- Built a full-stack MERN job posting platform with JWT-based authentication and bcrypt password hashing.
- Implemented job CRUD functionality with listings linked to user profiles via Mongoose relationships.
- Built the complete React frontend: registration/login, job browsing, personal job management, and protected routing.

HerMoney Landing Page

2025/2026

Stack: *React 19 • TypeScript • Vite 5 • Tailwind CSS*

- Built a conversion-focused landing page for a financial education brand supporting women in building money confidence.
- Implemented a multi-language toggle (NL/EN/TR/FR), scroll-triggered animations, and a fully mobile-responsive layout.
- Live: her-money-ui-six.vercel.app

Freelance: n8n Automation Projects

2025/2026

Stack: *n8n • OpenAI API • Apify • Google Sheets API • Telegram API • Webhooks*

- Apify Listings Processing & Alert Automation — n8n workflow fetching Apify listing data, scoring each item via OpenAI, filtering by score, storing approved listings in Google Sheets, and sending Telegram alerts automatically.

- Amazon Wine Recommendation Automation — n8n workflow searching Amazon.it by budget and request, verifying availability, filtering to top recommendations, and returning structured JSON via webhook.

Freelance: Vehicle Departure Prediction (Power BI)

2025

Stack: *Power BI • Python • Data Analysis*

- Built a predictive analysis identifying which drivers were likely to be late vs on-time for departures.
- Delivered a Power BI dashboard and a 5-page written report summarizing findings as the final client deliverable.

Vehicle Insurance Prediction — MLOps Pipeline

2026

Stack: *Python • scikit-learn • XGBoost • MLflow • Docker • AWS S3*

- Built a complete ML pipeline covering data preprocessing, feature engineering, model training, and evaluation.
- Implemented MLflow Model Registry, Docker containerization, and AWS S3 for artifact storage as part of structured MLOps practice.

SmartInfo — YouTube & Document Content Analyzer

Final Year Project, 2025

Stack: *Python • LangChain • HuggingFace Transformers • FAISS • OpenAI API • React.js • FastAPI • Streamlit*

- Designed an end-to-end RAG pipeline with FAISS vector search and LangChain agent tools for Q&A over YouTube transcripts and documents.
- Fine-tuned HuggingFace transformer models; integrated a React.js frontend with a FastAPI backend.

Computer Vision & ML Practice Projects

2023 – 2025

Stack: *Python • TensorFlow / Keras • PyTorch • scikit-learn • OpenCV*

- Handwritten Digit Classification — CNN-based classifier on handwritten digit image data.
- Garbage Detection & Classification — computer vision project detecting and classifying waste types from images.
- Stock Price Prediction — time-series ML project predicting stock price trends.

Personal Portfolio Website

2025/2026

Stack: *React • Vercel*

- Built and deployed personal portfolio showcasing projects, skills, and background. Live: jawad-ali-rose.vercel.app

SKILLS

- Professional: LangChain, LangGraph, Agentic AI, Chatbots, LLMs / Generative AI, FastAPI, Flask, JavaScript (Node.js / Express), React.js
- Automation Tools: Copilot Studio, n8n, Make.com, Webhook Orchestration, Event-Driven Workflows
- AI Models & APIs: OpenAI API, Claude API (Anthropic), HuggingFace Transformers, Google AI / Gemini, Mistral AI, Ollama
- Vector Databases & Search: FAISS, Pinecone, ChromaDB, Elasticsearch, OpenSearch
- Databases: PostgreSQL, MySQL, MongoDB, Redis, SQLite, SQL Server, Firebase
- MLOps & DevOps: MLflow, Docker, Kubernetes, CI/CD Pipelines, AWS (EC2, S3, Lambda), Azure, GCP, Git, GitHub
- ML & Deep Learning: scikit-learn, XGBoost, LightGBM, CatBoost, TensorFlow / Keras, PyTorch, Transfer Learning
- Data & NLP: Pandas, NumPy, NLTK, spaCy, Selenium, BeautifulSoup, OCR (Tesseract), Data Preprocessing, Feature Engineering
- Frontend & Demos: Streamlit, Gradio, React.js, MERN Stack
- Analytics & BI: Power BI, Data Analysis, Reporting

CERTIFICATIONS

Anthropic AI Courses

May 2026

Completed: Claude 101, Claude Code In Action, Claude With Anthropic API, Intro To Agent Skills, Intro to MCP